

RECRUITAI

PROJECT SPECIFICATION & IMPLEMENTATION BLUEPRINT

Mechanical Prototyping Management Platform Backend

Detailed project brief generated from the project record.

CANDIDATE

Karan Oberoi

DURATION

21 days

TECHNOLOGY STACK

Python

MySQL

REST API

TypeScript

React

Product Design

R&D

CAD/CAM

Prototyping

Project Overview

Summary of the problem, purpose and expected outcome.

PROJECT SUMMARY

Build a secure, high performance REST API that powers a platform for engineers to manage CAD prototypes, request manufacturing, and generate usage reports.

CANDIDATE Karan Oberoi	TYPE backend	DURATION 21 days
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Primary Objectives

- Implement a production grade REST API using Python and MySQL.
- Design a normalized relational schema covering users, prototypes, files, and requests.
- Secure the API with JWT authentication and role based access control.
- Provide automated unit, integration, and performance tests.
- Create Docker based CI/CD pipeline and deployment scripts.

Candidate Details

NAME

Karan Oberoi

Technology Stack

Technologies, frameworks and tools expected for implementation.

Python

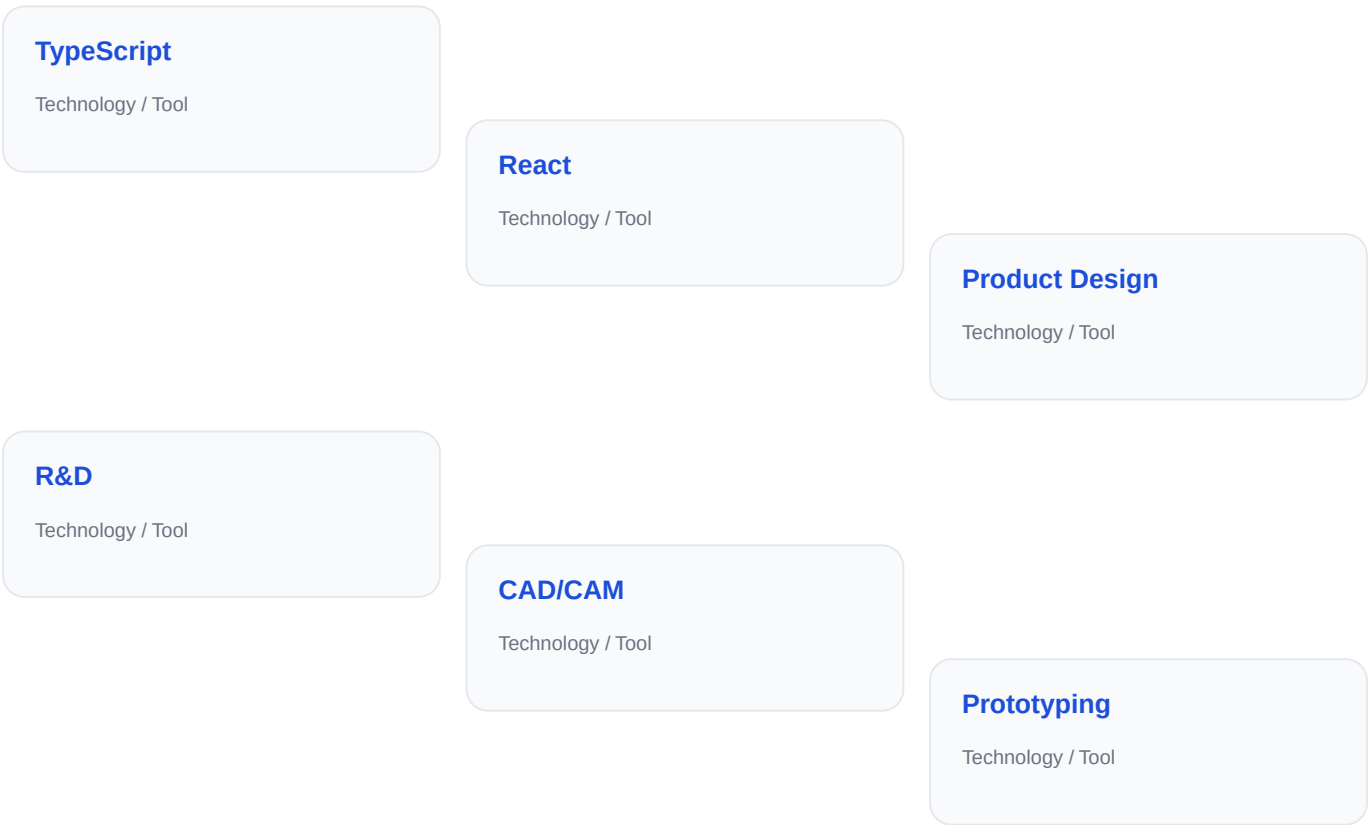
Technology / Tool

MySQL

Technology / Tool

REST API

Technology / Tool



Functional Requirements

- Engineers can create, read, update, and delete prototype projects with metadata.
- Users can upload and download CAD files (STEP/STL) linked to a prototype.
- Managers can submit manufacturing requests tied to a specific prototype.
- Admins can generate PDF reports summarizing prototype counts and pending requests.
- All actions are exposed via versioned REST endpoints.
- System sends email alerts when a manufacturing request is approved or rejected.

Non-Functional Requirements

- Passwords are hashed with bcrypt and never stored in plain text.
- Input validation must reject malformed data with HTTP 400 and error details.
- Support at least 10 concurrent file uploads without degradation.
- Read heavy endpoints must respond within 200 ms under typical load.
- Deployable via Docker with environment based configuration for DB credentials and JWT secret.

Modules & Features

Major functional areas that make up the project.

User Management

Handles registration, login, JWT issuance, and role assignments.

- Register/Login
- Role based access control

- Password reset

Prototype Management

CRUD operations for mechanical prototype projects and associated metadata.

- Create/Update/Delete prototypes
- Search and filter prototypes

CAD File Service

Secure storage, retrieval, and versioning of uploaded CAD files.

- File upload with virus scan
- Encrypted storage
- Download with permission check

Manufacturing Workflow

Process for managers to request manufacturing and track status.

- Submit request
- Status updates
- Email notifications

Reporting & Analytics

Generate usage statistics and exportable reports for admins.

- PDF report generation
- Dashboard data endpoints

Implementation Roadmap

Recommended phases and milestones for completing the project.

1

Foundation

Initialize git repo, Create Dockerfile and docker compose, Configure GitHub Actions for linting and testing

- Initialize git repo
- Create Dockerfile and docker compose
- Configure GitHub Actions for linting and testing

2

Core API & Database

Design ER diagram, Implement SQL migrations, Create FastAPI app with routers for users and prototypes

- Design ER diagram
- Implement SQL migrations
- Create FastAPI app with routers for users and prototypes

3

Authentication & Authorization

Integrate PyJWT, Add dependency for role based guards, Write auth middleware and tests

- Integrate PyJWT
- Add dependency for role based guards
- Write auth middleware and tests

4

CAD File Management

Choose storage backend (local volume), Implement upload endpoint with virus scan stub, Encrypt files at rest using Fernet

- Choose storage backend (local volume)
- Implement upload endpoint with virus scan stub
- Encrypt files at rest using Fernet

5

Manufacturing & Reporting

Create ManufacturingRequest model, Add email notification via SMTP mock, Integrate ReportLab for PDF generation

- Create ManufacturingRequest model
- Add email notification via SMTP mock
- Integrate ReportLab for PDF generation

6

Quality, Performance & Deployment

Write load test for file upload (Locust), Add caching headers for read endpoints, Configure production Docker compose with env vars

- Write load test for file upload (Locust)
- Add caching headers for read endpoints
- Configure production Docker compose with env vars

Implementation Note

This document is generated from the project record. AI-generated requirements should be reviewed by an administrator before implementation.